

REQUIREMENTS TABLE:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow the Sanitation Supervisor to add and update waste collection locations.	High	When valid location details are entered and submitted, the system shall save the location and display it in the collection location list.	Accurate collection locations are required for planning efficient waste collection routes.
FR-002	Functional	The system shall generate an optimized waste collection route based on the selected collection locations.	High	When the supervisor selects valid collection locations, the system shall generate and display an ordered route within 10 seconds.	An optimized route helps reduce travel distance, fuel consumption, and collection time.
FR-003	Functional	The system shall allow the Sanitation Supervisor to assign a generated route to a Truck Driver.	High	After a route is assigned to a driver, the assigned route shall be visible to that driver.	Ensures that each driver receives the correct waste collection route.
FR-004	Functional	The system shall allow the Truck Driver to view the assigned collection route.	High	The system shall display all assigned collection locations in the correct route order.	The driver needs the route to complete waste collection efficiently.
FR-005	Functional	The system shall allow the Truck Driver to update the collection status of each location.	Medium	When a location is marked as Completed or Not Completed, the updated status shall be saved and visible to the Sanitation Supervisor.	Enables the supervisor to monitor the progress of waste collection.
NFR-001	Nonfunctional	The system shall generate an optimized route within 10 seconds for up to 100 collection locations.	High	Testing with 100 collection locations shall show route generation completed in 10 seconds or less.	The system should provide route information quickly for efficient operations.
NFR-002	Nonfunctional	The system shall provide a user-friendly interface that allows users to access the main	Medium	During testing, a user shall be able to access any main function within a maximum of 3	A simple interface helps the supervisor and truck driver use

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		functions with minimal navigation.		navigation steps from the main screen.	the system efficiently.